KVL derivation for three-phase unbalanced system

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Introduction

In this document, I review two papers. The first papers introduces the derivation is for three-phase, balanced systems. The second derviation is for three-phase unbalanced systems.

First Paper: Network Reconfiguration in Distribution Systems for Loss Reduction and Demand Balancing, Baran et. al.

Paper's Objectives

1- How can we change the network configuration to reduce resistive losses? 2- How can the load be distributed such that no overloads occur on the T-Lines?

Equations Derivation

NOTE: The paper uses DistFlow equations, which we are going to use in this work.

Consider the following single line diagram:

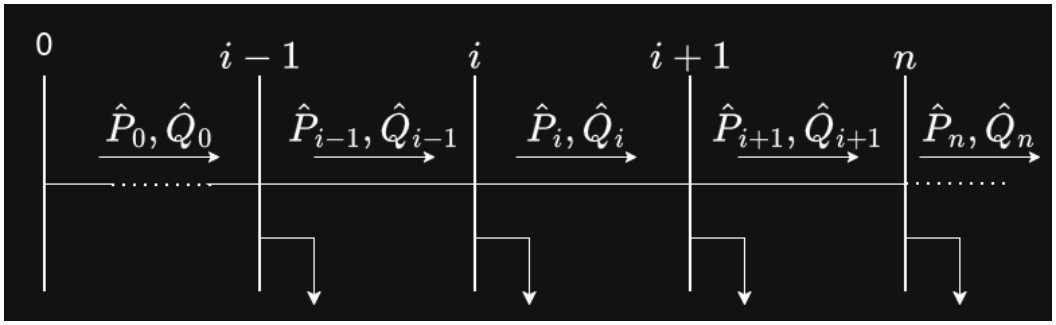


Figure 1: A single Line Diagram (SLD) for an n bus system. The loads are unknown for now. The Apparent Power ($\bar{S}$) flowing between each bus is shown.

For each load, we consider P_{L_n} and Q_{L_n}, indicating load power flow at each load.

Each line has an impedance, $Z_n = r_n + jx_n$. The apparent power flow at each line is given by:

$$\bar{S_n} = P_n + jQ_n$$

For instance, at bus $i+1$:

$$\bar{S}_{i+1} = P_{i+1} + jQ_{i+1}$$

I think it is easier to see this if we look at the power flow between two buses. Consider the power flow between buses 0 & :

Decomposing the Losses term in equation Equation 1

At any given bus :

Recall:

Wherein the magnitude of the apparent power is given by:

Squaring both sides:

Solving for the current, :

At any given bus , the current is:

substituting the current equation Equation 3 into Equation 2 we get:

Subtituting Equation 4 in the KCL between buses and in Equation 1, we get:

> **Note**
>
> Equation 5 is the real power DistFlow Equation. This equation can be used for any power flow calculations.

Similarly, the reactive power can be calculated in a similar manner:

Deriving the Voltage Equation:

Applying KVL to buses, and in Figure 1:

The current complex value can be solved for in a similar manner as in equation Equation 3:

Solving for :

Subtituting Equation 8 in Equation 7 & solving for :

We can then extend this to get only the magnitude of the voltage:

Recall :

Multiplying both terms, we end up with:

Recall

So the middle term can be decomposed such that:

Multiplying and obtaining only the real part because we are trying to calculate the magnitude:

The last term in Equation 10 can be simplified further. Recall that and

Subtituting Equation 11 and Equation 12 in Equation 10:

> **Note**
>
> Equation 13 is the Voltage Magnitude DistFlow Equation. This equation can be used for any power system.

Three-phase Unbalanced System

Using the same KVL we used before, Equation 9, we can set a system of equations for buses and such that:

For any given bus, the voltage equation can extracted from the above matrix form. For instance: